



SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
Mechanical
IA/IB/IC Schedule of 3Phase Electric Locomotive [Inspection]

लोको संख्या Loco No. 38458 क्लास Class IC शिड्यूलमें लेने का दिनांक Date taken under Sch 17/10/24

	आगमना Incoming / शिड्यूल Schedule		अंतिम निरीक्षण Final Inspection	
	अपर डैक Upper Deck	अंडर ट्रक Under Truck	अपर डैक Upper Deck	अंडर ट्रक Under Truck
Date/ Shift	17/10/24	22.10.24		
Name of Technician	Gulab Singh	Manoj + Manish		
Designation/ Token No.	mif			
Signature				

Customer Feed Back/Bookings (Record feedback given by Loco pilot)				
Sr. No.	Customer Feed Back/Bookings	Action Taken	Name of TCN / Attended By	Remarks, if any & Sign. of JE/SSE

Inspection Wing Mechanical: Check and record Following

	CPA Loading Time (Pressure 0 to 8.0 kg/cm ²)	Safety Valve Blowing	Pantograph-1		Pantograph-2		Compressor Loading Time (Pressure 8 to 9.5 kg/cm ²)		CP Safety Valve Blowing	Unloaded valve
			Raising	Lowering	Raising	Lowering	Comp-1	Comp-2		Operation time Approx. 12 Sec.
Std. Value		8.5 ± 0.25 kg/cm ²	6 to 10 Second	Less than 15 Second	6 to 10 Second	Less than 15 Second	120 Second	120 Second	11.5 ± 0.35 kg/cm ²	
GI	10.2	8.20	7.47	6.20	9.17	8.80	Sec	7.3 Sec	711.5	5.5 Sec
Final	10.5	8.5	7.5	6.5	8.5	9.0	Sec	6.8 Sec	711.5	6.5 Sec

	MR safety Valve pressure		Duplex check valve setting/Brake Pipe Pressure by Auto Brake (A9)		Feed Pipe Pressure		Brake Cylinder Pressure by Direct Brake (SA9)		Horn Sound			
	Blowing	Closing							Std. - Good			
	10.5 ± 0.2 kg/cm ²	9.5-9.8 kg/cm ²	5.0 ± 0.1 kg/cm ²		6.0 ± 0.2 kg/cm ²		3.5 ± 0.1 kg/cm ²		Cab-1		Cab-2	
Std. Value			Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2	LP	ALP	LP	ALP
GI	11.49	10.0	5.1	5.0	6.2	6.3	3.7	3.6	6.1	6.1	6.1	6.1
Final	10.7	9.8	5.1	5.0	6.0	6.2		3.5	6.1	6.1	6.1	6.1

	Computerized Control Brake (CCB) System if applicable		Air Flow	Air Dryer		ADC	MRC Time By both CP 0-10kg/cm ²
	Alarmer and Penalty	Working	Working	Working, Tower to change every minute (FTIL, SIL) & In every two minutes (KHL)	Humidity Indicator Colour Std-Blue	Working	3.5 Minutes (Max.) For 1750 LPM or [8.5 Minute for each 1450 LPM CP]
Std. Val.	Passed successfully	Working					
GI	E.70		10	working	blue	working	3 min 58 Sec
Final				working	blue	working	3 min 52 Sec

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	CPA Pressure build up time 8.5 kg/cm ²	BP drop in 5 Minute	FP drop in 5 Minute	MR drop in 15 Minute	Press BPPB to apply or release parking brake. Check Pressure in Parking Brake gauge		VCB Pressure switch setting
Std. Value	60 Sec. Max.	Max. drop 0.15 kg/cm ²	Max. drop 0.70 kg/cm ²	Max. drop 1.0 kg/cm ²	6.0 ± 0.15 kg/cm ²		Opens 4.5 ± 0.15 kg/cm ² , Closes 5.5 ± 0.15 kg/cm ²
GI	65 Sec	No Drop	No Drop	Drop 0.0 kg	Cab-1 N/A	Cab-2 N/A	Dummy
Final	66 Sec	No Drop	No Drop	2.0 kg			

	ACP on 4mm test plate		Auto brake controller Position	BP (kg/cm ²)		BC (kg/cm ²)	
				Cab-1	Cab-2	Cab-1	Cab-2
Std Values	Working		Run	5.0 ± 0.10	5.0	0.0	0.0
			Initial Application	4.6 ± 0.10	4.7	0.40 ± 0.10	0.8
	Cab-1	Cab-2	Full Service	3.35 ± 0.20	3.4	2.50 ± 0.10	1.8
			Emergency	Below 0.30	0.0	2.50 ± 0.10	1.8
Gen. Insp.	OK	OK				2.1	2.5
Final						2.5	3.5

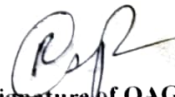
	Loco pilot cabin condition		Wiper Operation		Look out Glass		Fire Extinguisher	Sand box	Sander Operation		Cleaning of Loco
	Cab -1	Cab-2	Cab -1	Cab-2	Cab -1	Cab-2			Cab-1	Cab-2	
Std. Values	Good	Good	Working	Working	Clear	Clear	Sealed	Full	Working		Cleaned
Gen. Insp.	OK	OK	OK	OK	C	C	Cab-2 NLS	Full	OK	NLS	Dirty
Final											

क्र.सं. SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
(A) Pneumatic System (When Loco is energised check, attend and record)				
1	Check correctness of reading of BP, FP, MR, BC, AFI & RS gauges.	OK		
2	Check on dummy of BP/FP pipe.	OK		
3	Check the working of PTDC	N/A		
4	Check the proper functioning of all the gauges of the driver desk BP: (i) Normal (ii) with 7.5 mm hole BP should not drop below 4 kg/cm ² in 60 second. FP: (i) Normal (ii) with 5.5 mm hole FP should not drop below 4.5 kg/cm ² in 60 second. MR: (i) Mainline MR leakage without application of SA-9 Direct Brake (ii) Mainline MR leakage with application of SA-9 Direct Brake	0.7 Drop 0.1 Drop		
5	(RS drop) Panto line air leakage 0.70 kg/cm ² in 5 minutes.	No Drop		
6	Check working of Panto selection switch.	OK		
7	Position of isolating cock on pneumatic panel. (70,74,136 in open condition & 47 in close condition.)	Checked OK		
8	Drain moisture from MR and Inter Cooler.	Checked OK		
9	Check functioning of air dryer for purging, cycle time and change over with MR pressure of 8.0 kg/cm ² .	OK		
10	Check Position of the Air dryer cock (13,14 in service & 187 in isolated)			
11	Check air dryer charging when compressor is in working. (FTIL -120 sec/SIL-180 sec)	TRZ DENT		
12	Check and attend leakage of air dryer, compressor valves, Pneumatic valves, Pneumatic panels & pipelines.	OK		

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(Signature)

क्र.सं. SN	कार्यवाही कार्य /विवरण का निरीक्षण /(शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
B	BRAKE CONTROLLER (AUTO AND DIRECT)			
1	Check the functioning of driver's direct air brake & automatic train brake.	check ok	}	17/10/24
2	Check the free movement of the brake controller.	check ok		
3	Perform repeated application and release of brakes with both auto and direct brake controller from both cabs and ensure that BP & BC pressure build up & drop is proper on the gauge. Ensure by physically going to individual main and parking brake cylinder that their operation is smooth and without air leakage.	check ok		
C	CARBODY FILTER: Check condition and clean car body filter and refit.	cleaned	}	18/10/24
1	Thoroughly clean all roof and side body filters with water jet.	cleaned		
2	Check tightness of Roof joint clamp and Roof top bolt.	check ok		
3	Remove Louvers & Fly screen and clean it, Repair if found defective.	Repair		
	IC sch.			
4	Remove the tread plate and clean the drainage channels. Check and adjust, if necessary, the door-operating rod.	check ok		
H	LOOKOUT GLASS & SIDE WINDOWS			
1	Clean any built up dirt and debris from the channel below the cab-sliding window. Clear the sliding window channel drains hole in the door opening.			
2	Clean look out glass with detergent/water solution.			
I	WASHER / WIPER			
1	Check the operation of windscreen wipers using the manual handles. Rectify any fault found.	check ok	}	17/10/24
2	Check the condition of windscreen wiper blades. Replace if required.	check ok		
4	Check the water level in the reservoir at No.1 and No.-2 end. Top up if required. Check wipers for proper operation	Top up		
6	Check for damage of air and water pipes	check ok		
7	Examine all components for wear, looseness scoring and cracks, renew any defective components.	check ok		
8	Check seat-mounting points for integrity and ensure fixing or fully tightened.	check ok		
J	Key interlocking: Check all keys are present & the complete system is operational	check ok		
	(IC Sch)			
9	Check the wiper and parallel arm assembly for wear and damage. Replace the arm or any part if there is excessive wear or damage.	do.		
10	Clean the washer nozzle jet. (IC Sch)	do.		
K	Fire Extinguisher			
1	Ensure availability of fire extinguishers and ensure that it is not expired. CO2 I - 30/12/24 CO2 II 30/12/24	No EXP.	}	
2	Check the condition of CO2 gas cylinder, its regulator and pressure gauge in both cabs. Check the CO2 gas pressures. Refill if necessary	check ok		
3	Clean pipeline by compressed air and check that there should not be any leakage. kg/cm ²			
4	CO2 Gas pressure and date: Cab-1 96 Cab-2 94 kg/cm ²	check ok		
5	Clean fire detection unit pipeline air intake holes with soft non-metallic brush & vacuum cleaner.			


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क्र.सं. SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
L	OIL LEAKAGE IN MACHINE ROOM / CLEANING OF MACHINE ROOM.			
1	Visually inspect the machine room for any signs oil leakage from traction converter, transformer, D.C. Link, and series resonant circuit capacitors and other electrical equipment and carefully clean the same. (Caution : Electric supply room)	check on		
2	Clean the complete machine room with the air pressurized supply and vacuum cleaner to remove all the dust particles.	check on		
M	SAND GEAR SYSTEM			
	a) Check Sand box, cover gasket, Ejector and replace gasket if necessary.	check on		
	b) Check all sander pipe for alignment and check fasteners also.	check on		
	c) Check and ensure tightness of Sand box foundation bolt.	check on		
	d) Ensure sander pipe height 60mm above rail level.	check on		
	e) Refill sand in all sand boxes.	check on		
N	OIL COOLING UNIT CASING WITH RADIATOR			
1	Remove all dust, dirt and debris from the radiator chamber via the machine room access cover. (By using vacuum cleaner).	check on		
2	Visually check the conservator tank along with its gauge glass for any sign of crack / damage / oil leakage.	check on		
3	Measure air delivery before and after cleaning of radiator.	check on		
4	Measure vibrations of oil cooling casing using vibration analyzer and take appropriate action	check on		
5	Visually check the casing and support arrangement for oil pumps on the casing for any sign of crack or the damage	check on		
6	Visually check the oil cooler radiator for any oil leakage from VPTFP and VPSR and external damage from top to bottom.	check on		
	IC Sch			
7	a) Clean radiator with hot water jet and check that the radiator is not blocked.	check on		
8	b) Tighten the front foundation bolts of casing mounted on to the radiator and check intactness of jail and Z-clamp welding crack.	check on		

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UPPER DECK [Inspect, Attend and Record]			
SN	Description	CB	Remarks
1.	Loose, missing and broken components/parts.	✓	
2.	Clamping and condition of its welding/fastener.	✓	
3.	Condition and working of all pressure gauges.	✓	
4.	Working of all brake valves.	✓	
5.	Leakage of air (MR) CP.I. Head. Check (2019)	✓	
6.	Gauges, securing of loco pilot cabin.	✓	
7.	Operation of Wipers.	✓	
8.	Co-Action working of A9 Auto brake.	✓	
9.	Fire Extinguishers (prescribed Nos.)	✓	
10.	Check tightness of compressor foundation bolt & abnormal sound.	✓	
11.	check slackness of compressor fan and no vibration in comp.	✓	
12.	Check tightness of air dryer foundation bolt.	✓	
13.	Ensure Air dryer indicator blue colour	Blue	
14.	Ensure clamping of all pipe to avoid vibration and rubbing.	✓	
15.	Ensure fitment of all split pins	✓	
16.	Check tightness of all bolts of Loco Roof	✓	
17.	Check window of both cab, shutter window door & corridor doors	✓	

Signature of OAG JE/SSE

(Signature)

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]			
SN	Description	CB	Remarks
1.	Loose, missing and broken components/parts.	✓	
2.	Clamping and condition of its welding/fastener.	✓	
3.	Air leakage in bogie.	No Leakage	
4.	Overheated parts / Abnormal wear in parts.	✓	
5.	Check and Ensure Brake block and Wheel clearance >10mm	✓	

(Remarks if any & sign):

Signature and Name of Technician for Inspection Incoming with date & shift:

[Signature] 16 to 24
17/10/24

Signature and Name of JE/SSE for Inspection Incoming with date & shift:

[Signature]

Signature and Name of Technician for Inspection Final with date & shift:

Signature and Name of JE/SSE for Inspection Final with date & shift:

अन्य किसी भी जानकारी Any Other Information:

[Signature]
Signature of QAG SSE Incharge (Inspection)

[Signature]
Signature of QAG JE/SSE

SCHEDULE PROGRESS CHART

SN	Date	Shift	QC	Work Details	Name of JE/SSE	Sign of JE/SSE	Remarks
1	17/10/24	16/24		PRE-INSPECTION			
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
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20							
21							
22							
23							

Signature of QAG JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC SCHEDULE OF 3 PHASE ELECTRIC LOCOMOTIVE [UNDERFRAME]

लोको संख्या Loco No	32458	क्लास Class	शिड्यूल में लेने का दिनांक Date taken under Sch.
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UNDERFRAME IA/IB/IC SCHEDULE & RECORDS

A-Check and record Hoot Wear, Flange Wear & Tread Wear (Record Wheel Diameter in Schedule or After TT)									
Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear	Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear
1	1049.2	2	0.5	1.5	7	1035.9	3	1.5	1.5
2	1049.7	2	0.5	1.5	8	1036	2	0.5	1.5
3	1049.4	2	0.5	1	9	1037	2.5	0.5	1.5
4	1049.5	2	0.5	1.5	10	1036.8	1.5	0	1
5	1049.8	2.5	1	1.5	11	1035	3	1.5	1.5
6	1050.2	2.5	1	1.5	12	1035	2.5	0.5	1.5

B- Check and Record Vertical & lateral clearance (Primary & Secondary) as per TC/0082 (REV-1) dated 17/09/2015

Standard Value	Primary vertical clearance (Cab-1 & 2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
27 to 35 mm	40.50	37.80	34.00	30.00	32.70	37.70	34.25	36.00	34.40	31.00	34.40	37.70

Standard Value	Primary lateral clearance (Cab-1 & 2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
15 to 22 mm	15.90	12.50	16.80	12.90	14.70	12.90	14.60	13.30	14.20	13.40	14.30	15.60

Standard Value	Secondary Vertical clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame			
	CAB-1 LP Side		CAB-1 ALP Side	
32 to 60 mm	37.80		35.90	

Standard Value	Secondary lateral clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame			
	CAB-1 LP Side		CAB-1 ALP Side	
45 to 55 mm	48.30		47.50	

C- Check and record wheel gauge below Parameters: (IC Schedule) WHEEL GAUGE [Permissible limit 1596+3-0.5 mm, (1596±0.5 mm for RB/New loco)]

Axle No.	A	B	C	Average	Remarks
1					
2					
3					
4					
5					
6					

Signature and Name of Technician

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P1 110 L1 110 P2 94 22 102
 Cattle yard 170 162 168 160
 Side buffer 1100 1100 1100 1090

22/10/24
 15/10/24

Check and record below Parameters in Schedule or if TT is done:

Axle No.	Standard (mm)		Cab-1		Cab-2	
Location	Max	Min	R1	L1	R2	L2
Buffer Length	635	615	636	633	640	638
Buffer Height	1105	1030	1095	1090	1090	1075
Rail guard height	118	115	120	118	100	95
CBC Height	1105	1030	1095		1065	

STRIKER WEAR PLATE	New 8 mm, Condemn 2 mm	Cab-1 OK	Cab-2 OK
SHANK WEAR PLATE	New 6 mm, Condemn 1 mm	Cab-1 OK	Cab-2 OK

E- Measure D SHAKLE and write Coupling No.

CAB-1

CAB-2

Check and Record Gear case Oil Level:

TM	OIL FOUND	OIL TOP UP	TM	OIL FOUND	OIL TOP UP
1	Full	—	4	Full	—
2	Full	—	5	Full	—
3	Full	—	6	Full	—

	Hand Brake Operation	CB Coupler Operation & Condition		Transition Screw Coupling Operation & Condition		CBC Lock Pin		HP/FF Pipe	
Std. Value	Working	Normal		Normal		Intact		Intact	
		Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2
Gen. Insp.		OK	OK	OK	OK	OK	OK		
Final Insp.		OK	OK	OK	OK	OK	OK		

Brake Piston Travel :						Brake Shoe release	Standard 3/9/10 mm
Class of Loco	Standard value	Observed (mm)					
WAG7/WAP7	107 to 117 mm	Location-	1	2	3	4	
WAG9		Location-	110	112	111	115	OK

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
F	DRAFT GEAR AND COUPLERS		
1.	Liberally apply general-purpose grease to the face of the center buffer coupler at the end of the Locomotive	Chauhan	
2.	Check the intactness of the CBC bottom support plate bolts	Chauhan	
3.	Check intactness and functioning of CBC operating handle.	Chauhan	
4.	Check visually the striker casting of CBC assembly.	Chauhan	
5.	Check for working of CBC, Lock and unlock it, locking arrangement, working of TC	Chauhan	
6.	Check washer lock at knuckle pin bottom	Chauhan	
7.	Check gap between Striker and Draft gear for draft gear slackness	Chauhan	
	IC Sch.		
8.	Check the knuckles against crack by DPT	Chauhan	22/10/24
9.	Check knuckle by gauge no. 1&2 (130-135 mm by go no go gauge)	Chauhan	18/10/24
10.	Check knuckle nose wear by gauge no.3 go no go gauge	Chauhan	
G	BUFFERS		
1.	Check buffers for any crack visually of the flange and intactness of foundation bolts, check nuts and split pins (4 nos. each),	Chauhan	

Signature and Name of Technician

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Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
	2. Liberally apply general purpose grease to the face of the buffers at the ends of the Locomotive. Also smear grease on the buffer head plunger.	Chen	
	IC Sch.	Chen	
	3. Conduct RDPT of Buffers to detect crack flaws.	Chen	
	4. Check and measure the buffer height and buffer length.	Chen	
H	SUSPENSION		
	1. Examine the primary and secondary springs for signs of damage, cracks and broken end visually with torch light and tapping by a hammer & replace, if necessary	Chen	
	2. Examine different types of dampers for oil leakage and intactness of fasteners and end fittings	Chen	
	3. Check the condition of various Spheris block in the primary and secondary suspension arrangement and of various equipment mounted in the under frame like axle guide link, traction motor and its support arm, gear case support arm for their healthiness (it should not be crack) and replace, if necessary.	Chen	
	IC Sch.	Chen	
	4. Proper torque of all fasteners related to suspension arrangement and mounting of equipment in under frame	Chen	
	5. RDPT of all TM nose and its support arm.	Chen	
I	TRACTION BAR		
	1. Check the well being and presence of the safety sling	Chen	
	2. Check the push/pull rod pivot heads for damage	Chen	
	3. Examine the traction link assembly for signs of damage. Any defective item should be renewed	Chen	
	4. Check the fastener at the traction ends and those retaining the traction rod for security. If a fastener is loose, it must be renewed and tightened to the specified torque. Ensure intactness of the locking plate. SM1-241	Chen	
	5. RDPT of traction bar at end (flange) portion & V-Ring	Chen	
	6. Housing flange.	Chen	
	7. Examine limit chain and links for wear, corrosion or damage	Chen	
	8. Inspect the safety slings, pins and R clips for wear and damage.	Chen	
	9. Check Aclathan/Vulkollon ring against any crack or shifting	Chen	
	10. Check gap in housing flange and traction link flange	Chen	
	11. Torque modified bolt M20 X65 by 260Nm (SM1 241)	Chen	
	IC Sch.	Chen	
	12. RDPT of Pivot post for any crack	Chen	
J	BOGIE FRAME AND BRAKE GEAR		
i	BRAKE BLOCK		
	1. Inspect the brake blocks for wear. If it is below the minimum limit, replace the complete locomotive set brake block.	Chen	
	2. Check the brake block key security	Chen	
	3. Check the brake rigging for loose, damaged or worn out parts Rectify as required.	Chen	
	4. Examine the bogie frame, transom beam & damper base for signs of crack/damage	Chen	
	5. Check fitment of Palm Pull rod J Bracket, hanger lever and TM safety rope wire.	Chen	
	6. Inspect the rubber bumps for damage deterioration. Replace if required	Chen	
	7. Examine all bogies mounted equipment (Including earthing straps) for signs of damage and security of fastening	Chen	
	8. Inspect the brake rigging rotation points and component contact surfaces. Rectify any faults found.	Chen	
	9. Check the earthing shunts provided on bogie and body.	Chen	
	Check intactness or axle box cover bolts and sealing.	Chen	

21/10/24
18/10/24

Signature and Name of Technician

क्यूएजी जेडीएसएसई QAG JE/SSE

18/10/24

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
	10. Inspect brake gear for any worn out loose, damaged, hanging or dragging parts. Confirm intactness	Chen	
	11. Test for correct operation of the brake cylinder actuator unit.	Chen	
	12. Check the proper tightness of parking brake & service brake pneumatic pipes.	Chen	
	13. Check the external hitting marks & physical damage on the brake cylinder actuator unit.	Chen	
	14. Check the condition of parking brake cylinder bellows If damaged, replace them.	Chen	
	15. On parking brake cylinder pulling spindle & ring should be intact.	Chen	
	16. Check proper functioning of parking brake & service brake cylinder.	Chen	
	17. Check the operation (manual releasing) of the parking brakes	Chen	
	18. Check the bogie isolating cock, should be in proper position.	Chen	
	19. Check the wear of scrubber blocks, replace if required (WAP5only)	Chen	
	20. Maintenance of Traction Motor support plate and bogie nose to Prevent crack/breakage of Traction Motor support plate (Holder for Traction motor suspension) should be done as per RDSO's SMI no. RDSO/2011/EL/SMI/0269 (Rev. '0') dtd, 18.05.11	Chen	
	IC Sch.		
	21. Inspect the bogie body safety pin eyehole at the bogie frame for wear or cracks.	Chen	
	22. Maintain the vertical & lateral clearances (Primary & secondary) as per TC-082	Chen	
ii	WHEEL SET AND AXLE BOX		
	1. Examine all the wheels visually for following defects	Chen	
	a) Cracks: if any, requires replacement	Chen	
	b) Flats: Length 40mm or more requires re-profiling	Chen	
	c) Cavities: Isolated cavities of length 15mm or more or multiple cavities of length 10mm or more and less than 50mm apart require re-profile/replacement	Chen	21/05/11
	d) Metal builds up Height 1mm, or above and length 50mm or more requires re-profiling. For length less than 50mm, remove the build up with hand tools	Chen	18/10/11
	e) Scoring and Grooving: Circumferentially around the wheel tread due to action of brake blocks require re-profile/replacement	Chen	
	2. Grease Axle box after 300000 KM Engine Run	Chen	
	3. Check the temperature of axle boxes. The axle boxes which are running warm or reported to have higher temperature in comparison with others shall be attended on priority by physically opening of axle box cover and outer angle ring. The suitable maintenance shall be done either in-situ or by dismantling the axle box.	Chen	
	IC Sch.		
	4. Measure and record the wheel diameters and tread profile. Refer RDSO's Tech. Circular no. ELRS/TC/0083 (Rev. "0"),	Chen	
	5. Disconnect the electrical cable from the earth return unit. Remove the earth return unit from the axle. Examine the bearings and rear sealing position for signs of grease leakage or deterioration.	Chen	
	6. Check the axle for any cracks with the help of ultrasonic flaw detector. At the same time visual inspection should be done. (IC Sch Every 06 month)	Chen	
iii	GEAR CASE (WAP7/WAG9/WAG9H)		
	1. Externally clean the gear case & its gauge glass. Inspect for any signs of oil leakage and any external hits/ damage. Verify integrity of gauge glass protective cover and clean oil sight glass for visibility.	Chen	
	2. Check the oil level in the gauge glass and top up if required. Ensure intactness of the drain plug along with its sealing. Check its tightness.	Chen	
	3. Check for any breakage of bolt and replace	Chen	
	4. Check for intactness of all fasteners of gear case	Chen	
	5. Tighten all gear case fasteners at prescribed torque	Chen	

Signature and Name of Technician

क्यूजी जेई/एसएसई QAG-JE/SSe

18/10/11

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
6.	Check the gear case oil visually and replace if found dirty or dust full level in the gear case and fill to the requirement	Chen	
7.	Check Gear case oil for metal contamination	Chen	
8.	Visually examine for fractures, the tyre profile for wear, flat, and skidding marks	Chen	
9.	Check wheel root, flange and Tread wear & record Change all unserviceable Brake Blocks	Chen	
10.	Check condition and operation of Hand Brakes. Lubricate & adjust as necessary.	Chen	
11.	Adjust Brakes and lubricate union	Chen	
12.	Lubricate Brake Gearing, Slack Adjusters, Buffers, CB Coupler Assembly	Chen	
13.	Split the cotter pin, if new provided	Chen	
	IC Sch.	Chen	
14.	a) Axle box old grease to remove and New grease to fill, old grease sample to be sent Lab for metal content check	Chen	
	b) Lubricate suspension roller tube bearing.	Chen	
	c) Check Transition screw coupling long and short shackle bar thickness by Go-No Go gauge (39.5 mm snap 1/25 mm snap 8)	Chen	
	d) Carry out DFT of Buffer Foundation and DPT/MPT of CBC Knuckle clevis & Yoke. Check size of Clevis, Knuckle, coupling by Go-No Go Gauge as specified. As per SMI-213/MIG-76	Chen	

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action takes in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

SN	Description	CB No.	Remarks
1.	Measure and record buffer & rail guard height.	✓	
2.	If skidded, fill Wheel Sledding Complain Report Form, FRMB/22 or if Biased wheel wear, fill up Check last for Based Wheel Wear Cause Invest, FILM/24)	✓	
3.	Proper Fitment of Brake Block and as Key	✓	
4.	Rubbing of brake block with wheel in brake release position	✓	
5.	Movement of brake pull rod in brake applied position	✓	
6.	Brake Shoe and Brake hanger Pin/Bolts	✓	
7.	Brake piston movement.	✓	
8.	Support arrangement for brake cross tie bars	✓	
9.	CBC Drooping.	✓	
10.	Operation of both Transition screw couplings	✓	
11.	Operation of CBC, Operating Rod and Coupler lock Pin	✓	
12.	Operation of CBC Knuckles	✓	
13.	CBC retaining plate and support plate bolts	✓	
14.	Side buffers and their mounting bolts.	✓	
15.	Bolts of Cattle guards and back supports.	✓	
16.	Fitment of Rail Guard	✓	
17.	Gear case mounting bolts and 'C' clamp bolts	✓	
18.	Suspension bearing cap bolts and it's sealing.	✓	
19.	Crack in TM Suspension taper roller bearing housing	✓	

Signature and Name of Technician

क्यूजी जेई/एसएसई QAG-JE/SSE

After TT Reading

Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear	Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear
1	1046.0	0.5	0.0	0.5	7	1033.5	1.0	0.5	0.5
2	1046.1	1.5	0.5	1.0	8	1032.8	2.0	1.0	1.0
3	1046.0	1.0	0.0	0.5	9	1033.5	0.5	0.0	0.5
4	1046.3	1.0	0.5	0.5	10	1033.0	1.0	0.5	0.5
5	1046.0	0.5	0.5	0.5	11	1033.2	1.0	1.0	0.5
6	1046.4	1.0	0.5	0.5	12	1033.0	1.5	0.5	1.0

Axle No.	Cab-1		Cab-2	
Location	RI	LI	R2	L2
Buffer Length	635	632	640	635
Buffer Height	1090	1090	1090	1085
Rail guard height	110	112	106	107
CBC Height	C = 1070		H = 1075	



22/10/21

Signature and Name of Technician

क्यूएजी जेई/एसएसई QAG-JE/SSE



पश्चिम रेलवे
Western Railway

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC SCHEDULE OF 3PHASE ELECTRIC LOCOMOTIVE
[COMPRESSOR & PNEUMATIC SYSTEM]

लोको संख्या Loco No क्लास Class शिड्यूल में लेने का दिनांक Date taken under Sch

क्र.सं. SN	कार्यवाही कार्य / विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
A	मेन कंप्रेसर और बेबी कंप्रेसर Main Compressor and Baby Compressor			
1	Check the pressure build up timing of each compressor independently in pre-inspection and take needful action during inspection. [From 0 to 10 kg/cm ² i) With 1750LPM Comp-7 Minutes max. ii) With 1450LPM Comp- 8.5 Minutes max.]	CPI - 17 MIN CPII - 6 MIN	53 Sec 13.5 Sec	8 min 20 sec 6 min 21 sec
2	Feel the vibrations of the compressor in pre-inspection and taken needful action during inspection.	checked		
3	Check for any external damage / abnormal sound / proper functioning of the compressor.	checked		
4	Check the complete compressor support brackets for their healthiness.	checked		
5	Visually examine the compressor mounting resilient blocks and take needful action.	checked		17/10/16
6	Visually examine the various interconnecting pipelines and its support clamps of the compressors for their healthiness / airleakage.	checked		
7	Cleaning of input air filter and the primary oil filter.	checked		
8	Clean the breather assembly.	checked		
9	Check and Drain the inter cooler / after cooler assembly and also observe its discharge and take needful action.	checked OK		
10	Ensure healthiness / intactness of various protection guards of compressor.	checked OK		
11	Check the condition of the coupling between compressor & motor.	checked OK		
12	Visually check the compressor fan blade for their healthiness.	changed		
13	Check the oil level in the compressor and top up as per requirement.	checked OK		
14	Carry out the inspection of each type of compressor in line with recommendation of respective manufacturer.	checked OK		
15	Check compressor foundation base for crack (For under slung compressor)	checked OK		
B	Aux. Compressor (CPA)			
1	Clean the CPA externally. Clean Suction Air filter.	cleaned		
2	Check oil level and condition of oil and top up if required with the oil specified (Servo Press 150).	drain & top up		
3	Check tightness of all nut bolts, foundation bolts.	checked OK		
4	Measure and record time & pressure of the auxiliary compressor	checked		
C	Sch IC: i) CP all discharge valves to remove.	CP-1 all safety discharge Valve checked		
	ii) CP all discharge valves refit by overhauled	CP-2 H/V valve changed		
	iii) CP oil to be change.	changed		
	iv) Baby compressor/CPA oil to be change	changed		
	CPA	changed		
	CP-1	changed		
	CP-2	checked		
	v) Baby compressor/CPA safety valve (8.5 kg/cm ²) to be replace by overhauled			
D	AIR SUPPLY AND PNEUMATIC SYSTEM			
1	NRV and unloader valves fail due to ingress of Carbon which extracted from CP. So, one cycle dismantling, checking and cleaning of NRV and unloader to be ensured in all type of 3-phase locomotives.	checked		
E	AIR DRYER			
1	Inspect the air dryer for any damage and its mounting for any damage,	checked		

क्र.सं. SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
	crack in mounting plate, intactness of fasteners. Check for any hit mark on main reservoir and drain cock. Check for proper position of the air dryer angle cocks.	check ok		
2	Check the integrity of the welding of the hanging support.	check ok		
3	Check humidity indicators for blue color. White indications mean the air dryer is not operating effectively. Desiccant shall be changed, if needed.	check ok		
4	Replace desiccant if found to be contaminated with either water or oil.	check ok		
5	Testing of air dryer for proper functioning to be done as per Technical Circular no. RDSO/2011/EL/TC/0108, Rev.'0' dtd. 7.3.11 for testing of air dryers and its efficacy on electric locomotives.	check ok		
F	E-70 BRAKE SYSTEM (PNEUMATIC PANEL)			
1	Visually inspect the pneumatic panel noting any sign of air leakages and rectify accordingly.	check ok		
2	Clean the pneumatic panel externally.	check ok		
3	Check for correct position of the cut out/isolating cock.	check ok		
4	Check all the cable connections for looseness or deficiency of cleats. If any inspection section is to be informed (IC Sch)	check ok		
5	Ensure provisioning and proper functioning of RS gauge, Switch, lamp & other gauges on pneumatic panel.	check ok		
6	Check the sealing of connectors, flexible pipe, regulator, air tank etc.	check ok		
G	BRAKE ISOLATING COCKS: Check for correct operations of BP & FP angle cocks. Check the condition of BP/FP pipe, cock & isolating handle for any hit mark or crack. Check correct condition of Bogie Isolating cock.	check ok		
H	BRAKE HOSES: Examine the brake hoses at the ends of the locomotive to make sure that they are not chafed or damaged. Check coupling cocks for correct operation.	check ok		
I	AIR RESERVOIR			
1	Blow down and drain the reservoir until exhaust of complete water and oil. Check all drain cocks are closed on completion.	check ok		
2	Check for any hit mark on main reservoir & drain cock	check ok		
3	Ensure intactness / tightness of main reservoir foundation bolts.	check ok		
4	Check the rubber gasket at the inspection cover of air reservoir for any damage/leakage.	check ok		
5	Check the pressure regulator operates correctly	check ok		
J	THROTTLE VALVE (IC Sch): Overhaul the throttle valve and change the rubber parts.	check ok		
K	IN LINE FILTER (IC Sch): Remove the filter element and clean thoroughly.	check ok		
L	HORN & HORN SWITCH			
1	Check condition of horn handle for any damage. Check for proper operation of horn.	check ok		
2	Check the condition of horn boots and replace if required	check ok		
3	Remove the horn & horn switch and Overhaul. Replace the diaphragm of horn. (IC Sch)	check ok		
4	Check all angle cocks for proper operation	check ok		
5	Ensure proper colour code of all the angle cocks.	check ok		
6	Check the tightness of the brake electronics control cards holding screws.	check ok		
7	Check all the pipes for leakage or damage. Check all the clamps for any damage and looseness.	check ok		
M	वायुवीय प्रणाली PNEUMATIC SYSTEM			
1	MR Safety valve, ADC, Panto throttle valve (Only for conventional Pantograph) replace by Overhaul in IC Sch.			

Signature and Name of Technician

Signature and Name of JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
CHECK SHEET FOR 3PHASE HR PANTOGRAPH ASSEMBLY (IA/IB/IC)

लोको संख्या Loco No 32458 क्लास Class शिड्यूल में लेने का दिनांक Date taken under Sch

SCHEDULE: PANTOGRAPH SR.NO. /Make//Type: LOCATION:

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken		Remarks
			PT-1	PT-2	
1.	Ensure that wearing strips are without sharp edges and no gap between end strip and wearing strip.	Smooth and no gap	checked	checked	✓
2.	Ensure availability of bolts, nuts, lock nuts, spring, washer, split pin etc.	All intact	checked	checked	
3.	Check the rubber air pipe for any leakage	No leakage	checked	checked	
4.	Check the rubber air bellow for any leakage/damage/rubbing	No leakage	checked	checked	
5.	Check lower frame assembly for any crack and play	No crack	checked	checked	
6.	Check upper frame assembly for any crack and breakage	No crack	checked	checked	
7.	Check coupling rod assembly for crack and its fitting intactness.	No crack & intact	checked	checked	
8.	Check diagonal rod for any crack both end side. (by DPT)	No crack	checked	checked	
9.	Check plain bearing eye bush for freeness.	free	checked	checked	
10.	Check ball bearing friction of Base frame, Upper frame and Coupling rod.	ok	checked	checked	
11.	Check Cam wear on top of cable rope guide surface.	OK	checked	checked	
12.	Check condition of all shunts (not more than 5% strand broken)	OK	checked	checked	
13.	Check pantograph pan for any external hitting marks	No hitting marks	checked	checked	
14.	Verify all welded joint for cracks.	OK	checked	checked	
15.	Check the control box for any air leakage	No air leakage	checked	checked	
16.	Check the condition and operation of hydraulic damper.	Normal	checked	checked	
17.	Check Parallel guide bar for free function of Panto Pan Head	working	checked	checked	
18.	Ensure that the thickness of wearing strips above the condemning size	New – 39 mm Condemn – 26 mm	1. 35.28 2. 31.98	1. 37.58 2. 38.20	✓
19.	Check the static contact force of pantograph at 6.4 kg/cm ² MR pressure	70 N / 7Kg Load Test	ac	ac	
20.	Check the raising and lowering time of pantograph	6 to 15 second	ac	ac	
21.	Check the horizontality of panto pan with spirit level.	Level OK	checked	checked	✓
22.	Check the function of auto drop device (ADD)	Working			

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken		Remarks
			PT-1	PT-2	
23.	Check the function of over reach detector (ORD)	Working	check ok	check ok	Done
24.	Check air filter for air leakage or not draining.	No leakage	check ok	check ok	
25.	Check the all pneumatic valve (3/2 valve, throttle valve and safety valve) for any leakage and proper function.	No leakage	check ok	check ok	
26.	Check pressure regulator for any leakage and proper function.	No leakage	check ok	check ok	
27.	Check the tightness of all bolts and thread locking compound to be provided in bearing base bolts.	Provided	check ok	check ok	
28.	Examine the roofline conductors for defects/looseness at joints.	No defect	check ok	check ok	
29.	Check the roof for any foreign material such as wire / clothpieces.	No foreign material	check ok	check ok	
30.	Check tightness, cleanliness and wear-tear of shunt connection between Panto and Roof line.	Tight, clean	check ok	check ok	
31.	Check that the pan head locking pins springs are in place and secure. Check that the head moves freely on its mounting pins attached to the apex frame. If movement is not free, check that the head is not twisted, the side beam are not bent/twisted & mounting pins are not loose or bent.	OK	check ok	check ok	
32.	Check tightness of the flexible shunt connecting screws. When a loose screw is found the shunt should be disconnected from the head &/or the pantograph frame, the mating surface cleaned & the shunt reconnected with new screws. Check all shunts for frayed strands and renew any shunt in which 5% or more of the strands are frayed.	OK	check ok	check ok	
33.	IC Sch: Clean filter element, if found brownish then replace with new.	Clear			

Remarks if any:

Signature and Name of Technician

Signature and Name of JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC Schedule of 3 Phase Electric Locomotive [Heavy Section]

लोको संख्या Loco No क्लास Class शिड्यूल में लेने का दिनांक Date taken under Sch

Sr. No	Details of Work / Inspection (Schedule)	Action Taken	Name of TCN	Sign / Remarks
A	VCB			
1	Check insulator for crack, VCB roof bus bar, chips and flash marks. If damage in service, renew it	checked		
2	Clean insulator, other parts that are dirty with soft, clean & dry cloth. (Do not use cleaning products based on fluoridated or chlorate compounds or sodium meta silicate), apply Silicon Spray.	checked		
3	Check contact spring (Earthing contact)	checked		
4	Check moving contact lever of auxiliary switch and tension spring & its mounting bolts for breakage	checked		
5	Check the cables & lugs for broken cables, lugs looseness	checked		
6	Check the main bushing condition, its connection and foundation bolts and clamps.	checked		
7	Check the tightness of HV Connections (Torque 70Nm)	checked		
	IC Sch:			
8	a) Check for damage to connection of the earthing isolator, Cleaning & greasing, if necessary.	checked		
	b) Check the torque of VCB fixing screw (Torque 70Nm)	checked		
	c) Remove the bottom cover and check that the control air and electrical connection are tightened & Ensure that there is no leakage from pressure regulator	checked		
B	MAIN TRANSFORMER			
1	Read off the oil level on the gauge situated on the conservator. Top up the oil as necessary and check for any signs of leakage. Clean the gauge with dry cloth	checked		
2	Inspect the colour of the silica gel. If it is pink, replace with blue silica gel	checked		
3	Examine the flanges of the pipe couplings and flexible hose that link the transformer and conservator	checked		
4	Check the availability of hosepipe over the oil pipe compensator and check stoochi coupling pipes for proper layout/fitment.	checked		
5	Check the main Transformer and its protection cover for any damage, crack & oil leakage. Also refer instructions contained in RDSO's letter no. EL/3.2.1/3-Ph / TC-0076 for arresting oil leakages cases"	checked		
6	Check the deformity of Transformer drain cock protection cover guard if deformed replace it	checked		
7	Perform the sample test on transformer oil. Check specific value of BDV, DGA, moisture and acidity and condition monitoring of loco transformer by Dissolve Gas Analysis. (RDSO/SMI/138 & OEM Doc. Dt. 27th Nov, 1995)	checked		
8	Check visually the condition of transformer foundation bolt and Nylon lock nuts for proper locking.	checked		
9	Check the Ecable. Clean and visually inspect the insulator condition and Electrical connection of transformer and converter for crack & flashed mark and ensure red marking	checked		
10	Visually inspect for leakage, damage, burning/overheat signs and all fixing clamps condition of oil cooling metallic pipes, Prismatic level gauge, TFP bushing, HV bushing, Hose pipe etc.	checked		

Signature and Name of Technician

Signature and Name of JE/SSE

Sr. No	Details of Work / Inspection (Schedule)			Action Taken	Name of TCN	Sign / Remarks
C	EARTHING CONNECTION.					
	a) Check the earthing connection tightening torque 70Nm.					
	b) Connection of earthing system in 3-phase locomotives type WAP5, WAP7 & WAG9 (As per RDSO/2007/EL/SMI/248 Rev.'0' dated 22.11.2007)					
	c) Check Earthing shunt of Bogie to Body					
	Axle box 1-3		Axle box 2-4			
	Axle box 5-7		Axle box 6-8			
	d) Check Earthing shunt of Axle to Bogie					
	Axle box 2		Axle box 3			
	Axle box 6		Axle box 7			

F-R-B Booking

1. CPA safety MV blowing 10.2 kg/cm² at 8.5 PSI/cm²
2. CP-1 taken 2 min 20 sec to make 10.0 kg/cm².
3. Panto-2 raising time 5 sec

Signature and Name of Technician

Signature and Name of JE/SSE